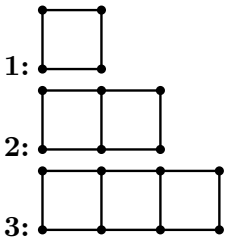
Linear Number Patterns – Extra Practice

These questions are extra practice in the same style as Questions 1 and 2 of the *Linear number patterns* worksheet: a picture pattern grows by the same amount each step, so the number of items follows a linear rule. For each question: (a) draw the next pattern, (b) complete the table, (c) find a formula, then (d)–(e) use your formula to answer questions about much larger patterns.

Question 1 – Matchstick Squares

This pattern shows the number of matchsticks needed to make a row of joined squares.



(a) Draw the next pattern (4 squares in a row).

Draw your pattern here.

(b) Copy and complete the table.

S	0	1	2	3	4	5	12
M							

Difference: ___ ___ ___ ___ ___

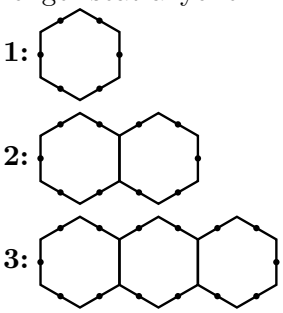
(c) Construct a formula for the number of matches (M) needed for any number of squares (S).

(d) If a row had 40 squares, how many matches would be needed?

(e) If a row had 75 squares, how many matches would be needed?

Question 2 – Hexagonal Picnic Tables

A campsite joins hexagonal picnic tables end to end in a row, as shown below (each dot is a seat). When two tables are pushed together, the shared side can no longer seat anyone.



(a) Draw the next pattern (4 hexagonal tables joined in a row).

Draw your pattern here.

(b) Copy and complete the table below.

T	0	1	2	3	4	5	25
C							

Difference: ___ ___ ___ ___ ___

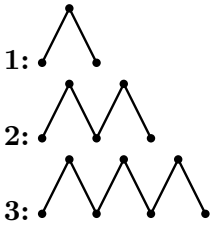
(c) Construct a formula for the number of seats (C) needed for any number of tables (T).

(d) If the campsite joined 40 tables in a row, how many seats would there be?

(e) If the campsite needed to seat 70 people in a single row of joined tables, how many tables would be needed?

Question 3 – Matchstick Mountains

This pattern shows the number of matchsticks needed to make a zigzag row of "mountain" peaks. Unlike the other patterns, the peaks don't share any matchsticks with each other – they just meet at a single point.



(a) Draw the next pattern (4 peaks in a row).

Draw your pattern here.

(b) Copy and complete the table.

N	0	1	2	3	4	5	20
M							

Difference: ___ ___ ___ ___ ___

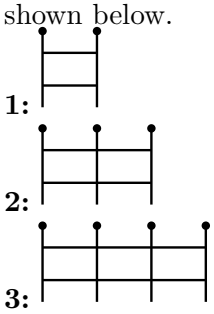
(c) Construct a formula for the number of matches (M) needed for any number of peaks (N).

(d) If a row had 50 peaks, how many matches would be needed?

(e) If a row had 80 peaks, how many matches would be needed?

Question 4 – Garden Fencing

A fence is built from posts with two horizontal rails running between each pair of neighbouring posts, as shown below.



(a) Draw the next pattern (5 posts).

Draw your pattern here.

(b) Copy and complete the table below.

P	0	1	2	3	4	5	6
R							

Difference: ___ ___ ___ ___ ___

(c) Construct a formula for the number of rails (R) needed for any number of posts (P).

(d) If a fence had 50 posts, how many rails would be needed?

(e) If a fence used 200 rails, how many posts would it have?